2nd Term 21/11/24 M/S

USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [23TCSE316]

THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

ESSENTIALS OF DATA WAREHOUSING AND DATA MINING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Why is ETL important in data warehousing?	L2	CO1	**02**
	b.	How are outliers identified in data cleaning?	L1	CO 2	**02**
	c.	What are the key goals of a data warehouse?	L1	CO 3	**02**
	d.	What is the role of the extraction phase in the ETL process?	L1	CO 4	**02**
	e.	What is classification in data mining?	L1	CO 5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	What is a Data Mart, and how does it differ from a Data Warehouse?	L1	CO 1	**05**
		OR			
	b.	Discuss the role of ETL (Extract, Transform, Load) in data warehousing.	L4	CO 1	**05**
3.	**a.**	Explain data quality service level agreement (SLA)	L5	CO 2	**05**
		OR			
	b.	Explain the data cleaning process.	L6	CO 2	**05**
4.	**a.**	What are the differences between OLTP and OLAP?	L2	CO 3	**05**
		OR			
	b.	What is Data Cleansing in the ETL process? Write a short note on it.	L4	CO 3	**05**
5.	**a.**	How do Data Warehouses support Business Intelligence? Discuss it.	L5	CO 4	**05**
		OR			
	b.	What are the key performance considerations in ETL processes?	L6	CO 4	**05**
6.	**a.**	Discuss the steps involved in the Knowledge Discovery in Databases (KDD) process.	L6	CO 5	**05**
		OR			
	b.	What are the benefits of Data Warehousing for organizations?	L2	CO 4	**05**
7.	**a.**	Explain the benefits of OLAP.	L4	CO 4	**05**

		OR			
	b.	Mention the key differences between single, two and three tier architectures.	L2	CO 3	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Discuss the importance of data integration in a data warehouse and the challenges involved.	L6	CO 1	**10**
		OR			
	b.	Differentiate OLTP and OLAP services and relate to the data warehouse.	L4	CO 1	**10**
9.	**a.**	Describe the role of Data Cleaning in the data pre-processing phase and the techniques used to handle noisy data.	L1	CO 2	**10**
		OR			
	b.	Discuss the role of data governance in an organization and how it impacts data management practices.	L6	CO 2	**10**
10.	**a.**	What is the architecture of a Data Warehouse? What are the challenges associated with Data Warehouse architecture?	L1	CO 3	**10**
		OR			
	b.	Explain in detail about ETL process in Data Warehousing.	L2	CO 3	**10**
11.	**a.**	Explain the process of information access and delivery process in data warehouse.	L2	CO 4	**10**
		OR			
	b.	Discuss about OLAP along with its operations.	L6	CO 4	**10**
12.	**a.**	Explain the Apriori algorithm for market basket analysis.	L4	CO 5	**10**
		OR			
	b.	Briefly discuss decision trees in data mining, and show how they make a decision.	L6	CO 5	**10**
13.	**a.**	Explain in detail about ETL. Explain their stages.	L4	CO 4	**10**
		OR			
	b.	Differentiate Data warehouse, Data lake, Data lake house.	L4	CO 2	**10**